

Introduction to Theoretical Computer Science

Lecture 12: The Polynomial Hierarchy, Alternation and **PSPACE**

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The Class **CoNP**

trick Question

Is the class **NP** closed under complement?

We don't know.

Question

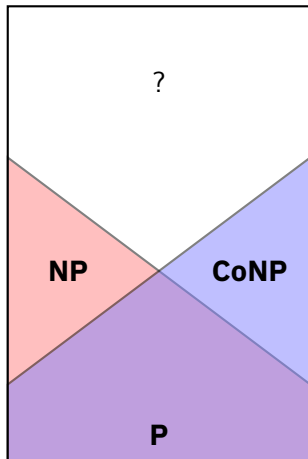
Why can't we just flip the answer, like we do for **P**?

Nondeterministic machines accept if **just one** path accepts. To flip the answer of an NRM, we'd have to accept an answer if **all paths** reject. This is no longer an (angelic) NRM (resp. NTM).

Question

What is **NP** $\cap$ **CoNP**? is it **P**? (we don't know)

What we have now



Sigmas

We shall introduce notation to describe polynomial problems.

Sigma

The set Σ_1^P describes all problems that can be phrased as $\{y \mid \exists^P x \in \mathbb{N}. R(x, y)\}$, where R is a **P-decidable** predicate and $\exists^P x \dots$ indicates that x is of size polynomial in the size of y .

- If a problem $Q \in \Sigma_1^P$ then Q is in **NP**. **Why?**
(we can “guess” an x and polynomially test $R(x, y)$)
- If a problem Q is in **NP** then $Q \in \Sigma_1^P$. **Why?**

Certificates

We can say that x is a **certificate** showing which “guesses” can be made by our NRM giving an accepting run.

So, **NP** = Σ_1^P .

Pi

The set Π_1^P describes all problems that can be phrased as $\{y \mid \forall^P x \in \mathbb{N}. R(x, y)\}$, where R is a **P-decidable** predicate and $\forall^P x \dots$ indicates that x is of size polynomial in the size of y .

$$\begin{aligned}
 \overline{\Sigma_1^P} &= \overline{\{x \mid \exists^P y. R(x, y)\}} \\
 &= \{x \mid \neg \exists^P y. R(x, y)\} \\
 &= \{x \mid \forall^P y. \neg R(x, y)\} \\
 &= \Pi_1^P
 \end{aligned}$$

As Σ_1^P is **NP**, Π_1^P is **CoNP**.

Deltas

For reasons that are unknown to me, some books have:

Delta

The set $\Delta_1^{\mathbf{P}}$ describes the intersection of $\Sigma_1^{\mathbf{P}}$ and $\Pi_1^{\mathbf{P}}$.

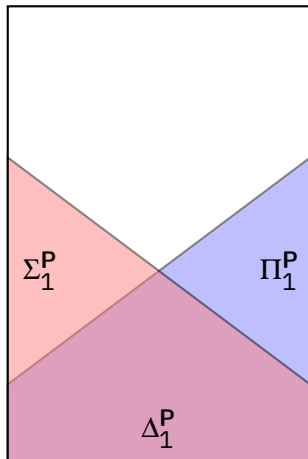
While others have

Delta

The set $\Delta_1^{\mathbf{P}}$ describes the set $\mathbf{P}$

From our characterisations of $\Sigma_1^{\mathbf{P}}$ and $\Pi_1^{\mathbf{P}}$, we have that $\Delta_1^{\mathbf{P}} \supseteq \mathbf{P}$, but we don't know if these definitions are equal.

Relabeling



Moving Higher

Definitions

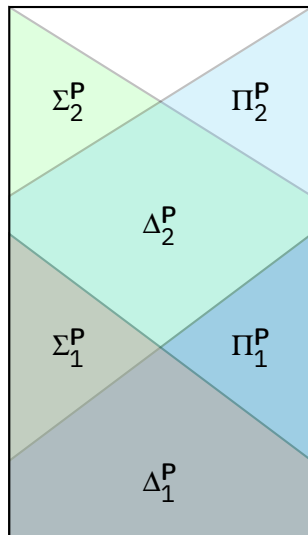
- Σ_2^P is all problems of form $\{x \mid \exists^P y. \forall^P z. R(x, y, z)\}$.
- Π_2^P is all problems of form $\{x \mid \forall^P y. \exists^P z. R(x, y, z)\}$.
- $\Delta_2^P = \Sigma_2^P \cap \Pi_2^P$

Note that $\Sigma_1^P, \Pi_1^P, \Delta_1^P$ are all $\subseteq \Delta_2^P$ (and therefore $\subseteq \Sigma_2^P$ and $\subseteq \Pi_2^P$).

Why?

(our R can simply “ignore” one of the parameters)

The Polynomial Hierarchy



An equivalent characterisation

We can define in terms of **oracles**:

- Δ_2^P is all problems that are decidable in polynomial time by some deterministic TM/RM with an $\mathcal{O}(1)$ oracle for some complete problem in Σ_1^P , i.e. it is **P** with an $\mathcal{O}(1)$ oracle for **NP**.
- Σ_2^P allows the TM/RM to be nondeterministic, i.e. it is **NP** with an $\mathcal{O}(1)$ oracle for **NP**.
- Π_2^P is **CoNP** with an oracle for **NP**.

Building up

In general, for any $n > 1$:

- Δ_n^P is all problems that are decidable by some deterministic, polynomially bounded TM/RM with an $\mathcal{O}(1)$ oracle for some problem $\in \Sigma_{n-1}^P$.
- Σ_n^P are all problems that are decidable by some **nondeterministic**, polynomially bounded TM/RM with an $\mathcal{O}(1)$ oracle for some problem $\in \Sigma_{n-1}^P$.
- Π_n^P are all problems decidable by some **co-nondeterministic**, polynomially bounded TM/RM with an $\mathcal{O}(1)$ oracle for some problem $\in \Sigma_{n-1}^P$.

Co-nondeterminism

Could also be called **demonic** nondeterminism. Like our normal (angelic) nondeterminism but only accepts if **all** paths accept.

Alternation

Alternation

Equivalently Σ_n^P are all problems that can be phrased as some **alternation** of (**P**-bounded) quantifiers, starting with $\exists^P$:

$$\{w \mid \exists^P x_1. \forall^P x_2. \exists^P x_3. \forall^P x_4. \dots x_n. R(w, x_1, \dots, x_n)\}$$

Π_n^P starts instead with $\forall^P$:

$$\{w \mid \forall^P x_1. \exists^P x_2. \forall^P x_3. \exists^P x_4. \dots x_n. R(w, x_1, \dots, x_n)\}$$

Alternating Machines

Alternating Machines combine the acceptance modes of both angelic and demonic nondeterministic machines.

Alternating Register Machines

Consider NRMs where instead of just a MAYBE instruction we have a $\text{MAYBE}^{\forall}$ instruction and a $\text{MAYBE}^{\exists}$ instruction.

- $\text{MAYBE}^{\exists}$ is a nondeterministic choice where we accept if **one** branch accepts.
- $\text{MAYBE}^{\forall}$ is a nondeterministic choice where we accept only when **both** branches accept.

Alternating Turing Machines are defined by labelling states with either $\forall$ or $\exists$.

Alternating Machines and the Polynomial Hierarchy

- The class Σ_n^P could equivalently be defined as the class of problems decided in polynomial time by an alternating machine that initially uses $\exists$ -nondeterminism, and every path in the machine swaps quantifiers (i.e. to $\forall$ or back to $\exists$) at most $n - 1$ times.
- Π_n^P is the same, except that we start with $\forall$ instead.

The class **AP**

AP is the class of all problems decidable by an alternating machine in polynomial time, without any restriction on swapping quantifiers.

AP is known to be equal to **PSPACE** (more on this in a moment)

A Fragile House of Cards

Warning

The polynomial hierarchy could collapse at any point.
(i.e. all of the classes in the PH could be equal)

We don't know that $\mathbf{P} \neq \mathbf{PSPACE}$, and the entire polynomial hierarchy is contained inside $\mathbf{AP}$ which = $\mathbf{PSPACE}$.

Wait, what's $\mathbf{PSPACE}$?

An RM/TM is *$f(n)$ -space-bounded* if it may use only $f(\text{inputsize})$ space. For TMs, space means cells on tape; for RMs, #bits in registers.

PSPACE is the class of problems solvable by polynomially-space-bounded machines.

The following inclusions follow from the definitions.

$\mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{PSPACE} \subseteq \mathbf{EXPTIME}$.

At least one of these inclusions must be strict, because we know that $\mathbf{P} \subset \mathbf{EXPTIME}$. But we don't know which one.

The common conjecture is that they are all strict.